WORKED SOLUTION

STEP I 1989, Question 1

Pure • Areas of Composite Regions

STEP I 1989, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q1

The key idea

The region is built from semicircles. You can get it by starting with one big half-disc, then taking away and adding smaller half-discs. So all we need is each radius, and in particular the radius of the lower semicircle CDE, which comes from the lengths along the line AG. Changing to parabolas makes no difference to the method, because a parabolic arch has a simple area formula that does the job of $\frac{1}{2}\pi a^2$.

Solution

The semicircles

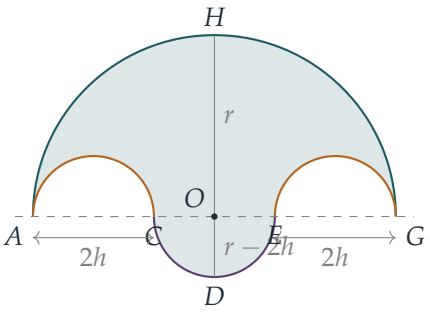
Measure along the line from A. Since ABC has radius h, $AC = 2h$. Similarly $EG = 2h$, and $AG = 2r$ because AHG has radius r. So

$$CE = 2r - 4h,$$

which is positive because $2h < r$. So the semicircle CDE has radius $r - 2h$. Its centre is the midpoint of CE, which is also the midpoint of AG by symmetry, so it shares its centre O with the big semicircle.

From the diagram, ABC and EFG lie on the same side of AG as H, inside the big semicircle, and CDE lies on the other side. The enclosed region is therefore

$$(\text{half-disc } AHG) \;-\; (\text{half-discs } ABC \text{ and } EFG) \;+\; (\text{half-disc } CDE).$$



A half-disc of radius a has area $\frac{1}{2}\pi a^2$, so the area enclosed is

$$\begin{aligned}\tfrac{1}{2}\pi r^2 - 2\cdot\tfrac{1}{2}\pi h^2 + \tfrac{1}{2}\pi(r-2h)^2 &= \tfrac{1}{2}\pi(r^2 - 2h^2 + r^2 - 4rh + 4h^2)\\ &= \tfrac{1}{2}\pi(2r^2 - 4rh + 2h^2)\\ &= \pi(r-h)^2.\end{aligned}$$

Ask yourself How long is HD?

The point H is the top of the big semicircle, so it lies on the perpendicular to AG through O, at distance r from O. The point D is the lowest point of CDE, so it lies on the same perpendicular, at distance $r - 2h$ from O on the other side of the line. So

$$HD = r + (r - 2h) = 2(r - h),$$

and a circle with diameter HD has radius $r - h$ and area $\pi(r-h)^2$. This is the area we found, as required.

The parabolas

Each semicircle of radius a is replaced by a parabolic arch on the same chord of length $2a$, with its vertex at height a above the midpoint of the chord. Put the origin at the midpoint of the chord. The parabola is symmetric about the axis through its vertex, so it has the form $y = a - kx^2$. It passes through $(\pm a, 0)$, so $k = 1/a$. The area between the arch and its chord is

$$\int_{-a}^{a}\left(a - \frac{x^2}{a}\right)\mathrm{d}x = 2a^2 - \frac{2a^3}{3a} = \frac{4a^2}{3}.$$

(This is Archimedes' result: two-thirds of the enclosing rectangle, $\frac{2}{3}\cdot 2a\cdot a$.)

Every arch still lies on the same side of AG as its semicircle did. We should also check that each small arch still lies under the big one. Both arches start from A with gradient 2, so this isn't obvious near A. At distance x from A, where $0 \le x \le 2h$, the big arch has height $x(2r - x)/r$ and the arch ABC has height $x(2h - x)/h$. Their difference is

$$\frac{x(2r-x)}{r} - \frac{x(2h-x)}{h} = x^2\left(\frac{1}{h} - \frac{1}{r}\right) \ge 0,$$

so ABC really does lie inside AHG, and EFG does too by symmetry. So the same add-and-subtract decomposition works, with $\frac{4}{3}a^2$ in place of $\frac{1}{2}\pi a^2$:

$$\begin{aligned}\tfrac{4}{3}r^2 - 2\cdot\tfrac{4}{3}h^2 + \tfrac{4}{3}(r-2h)^2 &= \tfrac{4}{3}(2r^2 - 4rh + 2h^2)\\ &= \tfrac{8}{3}(r-h)^2.\end{aligned}$$

So with parabolas the enclosed area is $\frac{8}{3}(r-h)^2$.

Checks

- Both answers are a constant times $(r-h)^2$. That has to happen, because the two calculations have exactly the same structure: only the factor for one arch changes, from $\frac{\pi}{2}$ to $\frac{4}{3}$. The ratio $\frac{8/3}{\pi} \approx 0.849$ is less than 1, as it should be, because a parabolic arch with the same chord and height lies inside the semicircle.
- Limiting case $h \to 0$: the small arches disappear and CDE becomes a full lower semicircle of radius r, giving a whole disc of area πr^2. The formula $\pi(r-h)^2$ agrees.
- Numerically, with $r = 3$ and $h = 1$ (the proportions of the paper's diagram), the areas are $4\pi \approx 12.566$ and $\frac{32}{3} \approx 10.667$. A shoelace computation on the traced boundary, for eight random pairs (r, h), matched both formulas to within the discretisation error.

Common slips

- Giving CDE radius $r-h$ or $r-4h$. Draw the lengths along AG: $CE = 2r - 4h$, so its radius is $r-2h$.
- Adding the small upper semicircles instead of subtracting them. They cut into the big half-disc, so their area comes off.

Beyond STEP

The figure in the first part is the *arbelos* ("shoemaker's knife"), a shape that Archimedes studied. Its classical form is a big semicircle with two smaller ones inside it, and its area also equals that of a circle on a well-chosen chord. Archimedes also found the $\frac{2}{3}$ rule for parabolic segments, by a limiting argument that comes close to integration. At university, the step from his "method of exhaustion" to the Riemann integral is a central part of a first analysis course. A good place to see it is Abbott, *Understanding Analysis*.

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